

Problem 2: Benchmarks, Baselines and Evaluation Metrics for Heterogeneous Graph Embedding

This answer covers graph embedding on *heterogeneous* graph structures for node classification (NC) and link prediction (LP). The homogeneous case is treated in Problem 1; homogeneous methods appear here only in their role as type-agnostic baselines.

1. Set-up: HINs, network schemas and meta-paths

Heterogeneous information network (HIN). Following Sun et al. (*PathSim*, PVLDB 4(11):992–1003, 2011) and the survey of Shi, Li, Zhang, Sun & Yu (*A Survey of Heterogeneous Information Network Analysis*, IEEE TKDE 29(1):17–37, 2017), a **heterogeneous information network** is a graph

$$G = (V, E, \phi, \psi), \quad \phi : V \rightarrow \mathcal{A}, \quad \psi : E \rightarrow \mathcal{R},$$

in which every node $v \in V$ carries exactly one *node type* $\phi(v)$ from the type set $\mathcal{A}$, every edge $e \in E$ carries exactly one *relation type* $\psi(e)$ from $\mathcal{R}$, and

$$|\mathcal{A}| + |\mathcal{R}| > 2$$

This inequality *is* the definition of heterogeneity: when $|\mathcal{A}| = |\mathcal{R}| = 1$ the network degenerates to the homogeneous graph of Problem 1. Each relation is typed at both ends, inducing a map $A_i \xrightarrow{R} A_j$ between two specific node types; its inverse R^{-1} is in general a *different* relation (*writes* vs. *written-by*), which is why the released benchmarks below count 6 or 8 “edge types” where a reader might expect 3 or 4.

Network schema. The **network schema** $T_G = (\mathcal{A}, \mathcal{R})$ is the directed graph whose vertices are the node types $\mathcal{A}$ and whose edges are the relation types $\mathcal{R}$. It is a meta-template of which G is an instantiation: every node and edge of G maps to some type in T_G , so the schema states which kinds of connection the dataset can express at all. This is the object Section 3 draws for each benchmark. Note that the drawing carries real information here whereas it does not in the homogeneous case: Cora’s schema is a single node type with one self-relation, while DBLP’s is a four-type star.

Meta-path. A **meta-path** $\mathcal{P}$ of length ℓ is a path defined *on the schema*,

$$\mathcal{P} : A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \dots \xrightarrow{R_\ell} A_{\ell+1},$$

denoting the composite relation $R = R_1 \circ R_2 \circ \dots \circ R_\ell$ from type A_1 to type $A_{\ell+1}$. A node sequence $v_1 v_2 \dots v_{\ell+1}$ in G with $\phi(v_i) = A_i$ and $\psi(v_i v_{i+1}) = R_i$ is a *path instance*. Meta-paths are written by type initials, so on DBLP APA abbreviates Author $\xrightarrow{\text{writes}}$ Paper $\xrightarrow{\text{written-by}}$ Author.

Meta-paths matter because each defines a *different* homogeneous projection of the same HIN, and the projections carry different semantics: on DBLP APA means *co-authorship* while APCPA means *published at the same venue*, and two authors may be strongly related under one and unrelated under the other. Nearly every method in Section 4 is a different answer to “how should meta-path structure be used?” — fixed and hand-chosen (metapath2vec, HAN), aggregated with attention (MAGNN), learned (GTN), or discarded in favour of per-relation parameters (R-GCN, HGT, simple-HGN).

2. Benchmarks

2.1 The standardised suite: HGB

Heterogeneous GNN papers before 2021 each preprocessed DBLP/ACM/IMDB their own way, so their reported numbers were not comparable. **HGB** (Heterogeneous Graph Benchmark; Lv, Ding, Liu et al., KDD 2021) exists precisely to fix this: fixed splits, fixed features, one leaderboard. Its headline statistics are Table 1; the per-type breakdown is Table 2.

Table 1: HGB (Lv et al., KDD 2021) benchmark statistics, as published.

Task	Dataset	#Nodes	#N. types	#Edges	#E. types	Target	#Classes
NC	DBLP	26,128	4	239,566	6	author	4
	IMDB	21,420	4	86,642	6	movie	5 [†]
	ACM	10,942	4	547,872	8	paper	3
	Freebase	180,098	8	1,057,688	36	book	7
LP	Amazon	10,099	1	148,659	2	product-product	—
	LastFM	20,612	3	141,521	3	user-artist	—
	PubMed	63,109	4	244,986	10	disease-disease	—

[†] **multi-label** — see Section 5.

Table 2: Per-node-type and per-relation breakdown of the four HGB node-classification benchmarks, recomputed from the released data files (see the note below).

Dataset	Node type	Count	Feat. dim	Relation	#Edges
DBLP	author *	4,057	334	author-paper	19,645
	paper	14,328	4,231	paper-term	85,810
	term	7,723	50	paper-venue	14,328
	venue	20	—	(+3 inverse relations)	
IMDB	movie *	4,932	3,489	movie-director	4,932
	director	2,393	3,341	movie-actor	14,779
	actor	6,124	3,341	movie-keyword	23,610
	keyword	7,971	—	(+3 inverse relations)	
ACM	paper *	3,025	1,902	paper-cite-paper	5,343
	author	5,959	1,902	paper-ref-paper	5,343
	subject	56	1,902	paper-author	9,949
	term	1,902	—	paper-subject	3,025
				paper-term	255,619
Freebase				(+3 inverse relations)	
	book *	40,402	film 19,427	music 82,351	
	sports	1,025	people 17,641	location 9,368	
	organization	2,731	business 7,153	no features at all	
	36 typed relations (BOOK-AND-BOOK 202,674, MUSIC-AND-MUSIC 283,670, ...)				

* labelled target type. Labelled counts: DBLP 4,057/4,057 authors; ACM 3,025/3,025 papers; IMDB 4,573/4,932 movies (of which **2,684 carry more than one label**); Freebase 7,954/40,402 books. Split is 24% train / 6% validation / 70% test on the labelled target nodes.

Provenance of Table 2. The HGB paper publishes only the totals of Table 1; it never prints a per-type breakdown. The numbers above were therefore recomputed directly from the released

`node.dat/link.dat/label.dat` files, and every one of the four datasets reproduces its published totals exactly (nodes, edges, number of node types, number of edge types and number of classes all match). That agreement is what licenses the table. Two incidental findings: HGB’s own `info.dat` declares the DBLP author attribute dimension as 0, but the data file actually carries 334-dimensional author features — and 334 is exactly the feature dimension HAN publishes for DBLP, which independently corroborates the data over the metadata; and ACM’s 1,902-dimensional bag-of-words is indexed by exactly the 1,902 `term` nodes.

2.2 The individual benchmark graphs

DBLP (bibliographic). A computer-science bibliography extract with types *author*, *paper*, *term*, *venue/conference*. Authors are labelled with one of 4 research areas (database, data mining, AI, information retrieval); author features are a bag-of-words over the keywords of the papers they wrote. It is the canonical author-classification HIN and the standard testbed for APA vs. APCPA semantics.

ACM (bibliographic). Papers from ACM venues with types *paper*, *author*, *subject*, *term*. Papers are labelled with 3 classes (database, wireless communication, data mining) and carry a bag-of-words feature. Distinctively, ACM has a *paper–paper* citation relation in addition to the star structure, and HGB splits it into a citation and a reference direction — so the schema is not a pure star.

IMDB (movie). Types *movie*, *director*, *actor* and (in HGB) *keyword*. Movies are labelled by genre. This is the benchmark where versions diverge most sharply: MAGNN’s IMDB has 3 *disjoint* genres (Action/Comedy/Drama), whereas HGB’s has 5 *overlapping* genres, making it multi-label.

Freebase. *The name denotes two unrelated graphs.* HGB’s Freebase is a general knowledge graph: 8 node types, 36 relations, 180,098 nodes, target type BOOK with 7 classes, and **no node features whatsoever** — so it isolates how much a model can extract from pure typed structure. It is book-centric only in the sense that BOOK is the labelled type; its schema is not a star but is in fact *complete* (Figure 2). HeCo’s “Freebase” is instead a *movie* dataset (3,492 movies, 33,401 actors, 2,502 directors, 4,459 writers; target movie, 3 classes). Quoting a Freebase number without saying which one is meaningless.

AMiner (bibliographic). Two scales. metapath2vec’s AMiner is web-scale: 1,693,531 authors, 3,194,405 papers, 3,883 venues, with 8 Google-Scholar categories as labels and 246,678 labelled authors. HeCo’s AMiner is a small subset (6,564 papers, 13,329 authors, 35,890 references; target paper, 4 classes). *Erratum worth knowing:* metapath2vec’s body text says the dataset “consists of 9,323,739 computer scientists”, but the authors’ own correction note states that 9,323,739 is the paper–author *link* count and the author count is 1,693,531. The wrong figure is still widely re-quoted.

Yelp and Douban (business / review). HERec’s Yelp has types *user* (16,239), *business* (14,284), *city* (47), *category* (511) and *compliment* (11), with U–B 198,397, U–U 158,590, U–Co 76,875, B–Ci 14,267 and B–Ca 40,009 edges (density 0.08%). Douban Movie has *user* (13,367), *movie* (12,677), *director* (2,449), *actor* (6,311) and *type* (38). Both are used for rating prediction / recommendation, reported with MAE and RMSE rather than F1.

Amazon (e-commerce) — heterogeneous in $\mathcal{R}$ only. HGB’s Amazon is the limiting case of the definition in Section 1, and worth stating for that reason. It has **one** node type (*product*, 10,099 nodes) and **two** relation types, *co-viewing* and *co-purchasing*, over 148,659 edges — the electronics subset preprocessed by GATNE (Cen et al., KDD 2019). So $|\mathcal{A}| = 1$ but $|\mathcal{R}| = 2$, giving $|\mathcal{A}| + |\mathcal{R}| = 3 > 2$: it is a heterogeneous network even though every node is the same type, which is precisely why the condition in Section 1 is stated as a sum rather than as “more than one node type”. Its network schema is therefore a single node carrying two distinct self-relations. It is used for **link prediction only** (it has no node labels), scored with ROC-AUC and MRR. It is also the cleanest illustration of the limit of meta-path methods: with one node type there is no type alternation to define a meta-path over, and HGB’s link-prediction table accordingly leaves MAGNN’s Amazon entry blank, noting that vacancies are “due to lack of meta-paths on those datasets”. Relation-aware but meta-path-free models (R-GCN, GATNE, simple-HGN) are the ones that can run on it at all.

LastFM and MovieLens (user–item HINs). MAGNN’s Last.fm has *user* (1,892), *artist* (17,632) and *tag* (1,088), with U–U 12,717, U–A 92,834 and A–T 23,253; it carries *no labels and no features*, so it is a pure link-prediction benchmark (predict user–artist), with one-hot IDs as dummy inputs. MovieLens/ML-100K is used analogously as a user–item bipartite HIN enriched with occupation/genre types. **PubMed**, in its HGB heterogeneous form, is a 4-type biomedical graph (63,109 nodes, 10 relation types) used for disease–disease link prediction — not to be confused with the homogeneous PubMed citation network of Problem 1.

OGB: ogbn-mag and MAG240M. These supply the scale the older benchmarks lack. **ogbn-mag** is a Microsoft Academic Graph subset with *paper* (736,389), *author* (1,134,649), *institution* (8,740) and *field of study* (59,965) — 1,939,743 nodes — and four relations: *author–affiliated_with–institution* (1,043,998), *author–writes–paper* (7,145,660), *paper–cites–paper* (5,416,271) and *paper–has_topic–field_of_study* (7,505,078), totalling 21,111,007 edges. The task is predicting each paper’s **venue** among **349 classes**; only *paper* nodes have features (128-d word2vec), the rest are featureless. Crucially the split is **temporal** — train ≤ 2017 , validation 2018, test ≥ 2019 — not random, so it measures generalisation forward in time. The official metric is accuracy. **MAG240M** (OGB-LSC) is the same design two orders of magnitude larger: 121,751,666 papers, 122,383,112 authors and 25,721 institutions (244,160,499 nodes) with 1,728,364,232 edges, 768-d RoBERTa features, and 153 arXiv subject classes.

2.3 Version incompatibility — the caveat that matters

The same dataset name denotes different graphs in different papers. Table 3 makes this concrete; quoting a statistic without naming the version is an error.

Four specific discrepancies are worth flagging.

1. **DBLP’s term count.** HAN reports 8,789 terms; MAGNN, HeCo and HGB all use 7,723. HGB’s published total is only consistent with the latter ($4,057 + 14,328 + 7,723 + 20 = 26,128$, whereas the HAN figure would give 27,194). HAN’s own table is also internally inconsistent: it lists 14,328 papers in the Paper–Author and Paper–Conference rows but 14,327 in the Paper–Term row.
2. **HGB’s ACM does not match its stated provenance.** HGB says it uses HAN’s subset while “preserving all edges including paper citations and references”, but its author count (5,959 vs. 5,835) and P–A edge count (9,949 vs. 9,744) differ from HAN’s published table.

Table 3: The same dataset name denotes different graphs in different papers. Figures that differ from the majority version are bolded.

Dataset	Version	Node counts	Note
DBLP	HAN	paper 14,328 / 14,327, author 4,057, term 8,789 , venue 20	P-T = 88,420
	MAGNN / HeCo / HGB	paper 14,328, author 4,057, term 7,723 , venue 20	P-T = 85,810
	GTN	18,405 total (= paper + author + venue; <i>term dropped</i>)	4 edge types
ACM	HAN	paper 3,025, author 5,835, subject 56	P-A = 9,744
	HGB	paper 3,025, author 5,959 , subject 56, term 1,902	P-A = 9,949
	HeCo	paper 4,019 , author 7,167 , subject 60	P-A = 13,407
IMDB	HAN	movie 4,780, director 2,269, actor 5,841	
	MAGNN	movie 4,278, director 2,081, actor 5,257	3 classes, single-label
	HGB	movie 4,932, director 2,393, actor 6,124, key-word 7,971	5 classes, multi-label

Re-adding citation edges cannot change the number of *authors*, so the two ACM graphs genuinely differ.

3. **IMDB changes task type, not just size.** HGB’s IMDB is multi-label with 5 overlapping genres; MAGNN’s is single-label with 3 disjoint ones. An F1 on one is not comparable to an F1 on the other even in principle (Section 5).
4. **HGB’s LastFM edge count double-counts one relation.** Its node total 20,612 exactly equals MAGNN’s $1,892 + 17,632 + 1,088$, but its edge total is 141,521 against MAGNN’s 128,804. The gap is exactly 12,717, the U-U count: $2 \times 12,717 + 92,834 + 23,253 = 141,521$. So the symmetric user-user relation is stored in both directions while U-A and A-T are stored in one.

3. Network schemas and meta-paths

Each schema in Figures 1 and 2 is $T_G = (\mathcal{A}, \mathcal{R})$ for one benchmark family: circles are node types (letter = the initial used in meta-path notation), edges are relation types. Inverse relations are omitted for readability in Figure 1; Freebase (Figure 2) has no inverse relations at all and is drawn complete.

A schema alone does not say which compositions of relations the literature actually uses. The meta-paths conventionally instantiated on each of these schemas, together with the semantic similarity each one encodes, are collected in Table 4; these are the paths that HAN, MAGNN and HERec take as input, and reading them against the schemas above shows how a two-hop alternation such as $A-P-A$ turns a typed bipartite relation into a homogeneous similarity over the target type.

Table 4: Meta-paths commonly used on each benchmark, and the semantic relation each encodes.

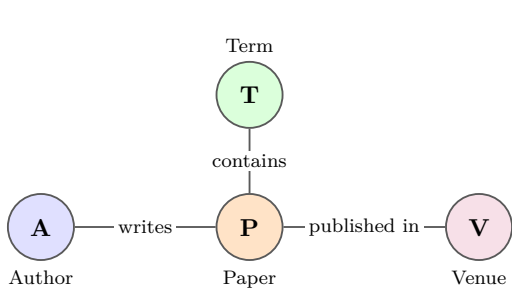
Dataset	Meta-paths	Semantics
DBLP	APA	two authors co-wrote a paper (co-authorship)
	APCPA / APVPA	two authors published at the same venue (community)
	APTPA	two authors used the same term (topical similarity)
ACM	PAP	two papers share an author
	PSP	two papers share a subject
	PcPAP, PrPSP	HGB variants routed through a citation (c) or reference (r) hop
IMDB	MAM	two movies share an actor
	MDM	two movies share a director
	AMA, DMD	two actors co-starred / two directors share a movie
ogbn-mag	PAP	two papers share an author
	PAIAP	two papers from the same institution (via author-institution)
LastFM	UAU	two users listen to the same artist
	UATAU	two users like artists sharing a tag
	AUA, ATA	two artists share a listener / a tag
Yelp	UBU	two users reviewed the same business
	UBCaBU	two users reviewed businesses of the same category
Freebase	BB, BFB, BPB	book-book; books sharing a film / a person

The remaining benchmarks described in Section 2 are drawn in Figure 3. Two of them are instructive rather than merely additional. **Amazon** is the degenerate case that justifies writing the condition of Section 1 as $|\mathcal{A}| + |\mathcal{R}| > 2$ rather than “more than one node type”: its schema is a *single* circle carrying two distinct self-relations, so it is heterogeneous in $\mathcal{R}$ alone. It is also why no meta-path can be defined on it — with one node type there is no type alternation to alternate over, which is the structural reason the meta-path-based models have a blank entry on it. **PubMed (HGB)** is the opposite extreme among the link-prediction sets, and is the second benchmark here whose schema is *complete*: all $\binom{4}{2} = 6$ unordered type pairs carry a relation and all four types carry a self-relation, $6 + 4 = 10$, matching its published count of 10 relation types. Its prediction target, DISEASE–DISEASE, is itself one of those self-relations — so unlike DBLP or ACM the task is to predict edges *within* a single type, which is why the type-alternation machinery of meta-paths buys so little on it.

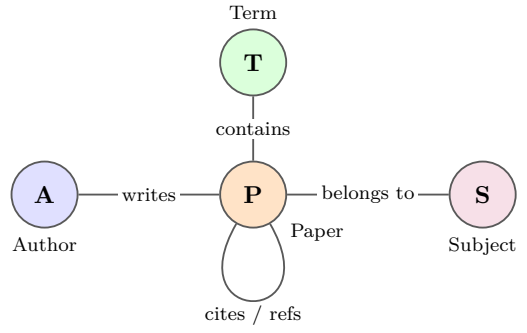
4. Baselines

4.1 Homogeneous methods, reported as type-agnostic baselines

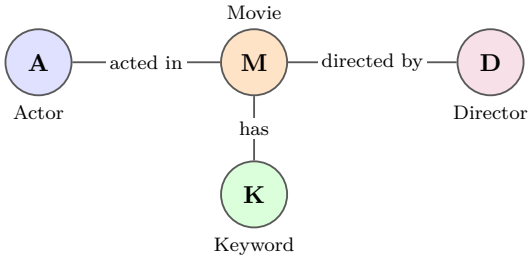
These are reported in essentially every heterogeneous paper, and the *point* of reporting them is that they **discard ϕ and ψ entirely** — they run on the graph with all type information erased (or on one meta-path-induced projection). They are the control condition: whatever a heterogeneous model gains over them is the measured value of modelling types. HGB’s central and uncomfortable finding is that this gap is much smaller than the literature had claimed, and that a properly tuned GAT beats most specialised HGNNs.



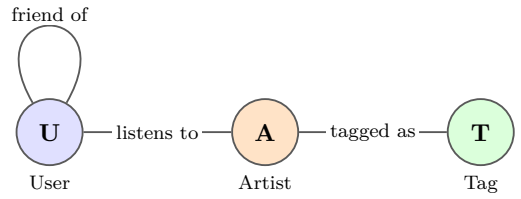
DBLP



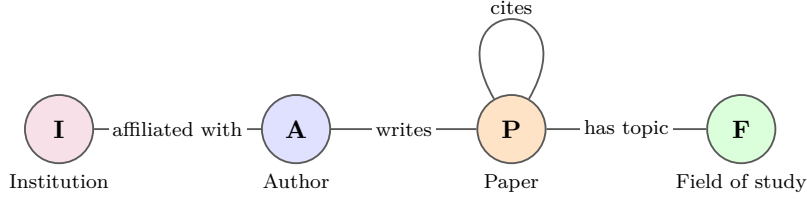
ACM



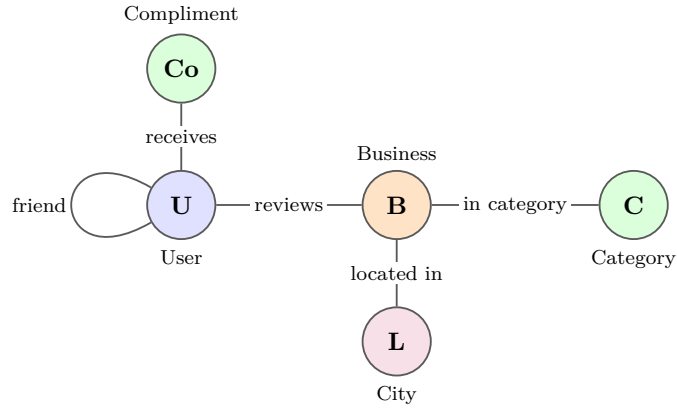
IMDB



LastFM

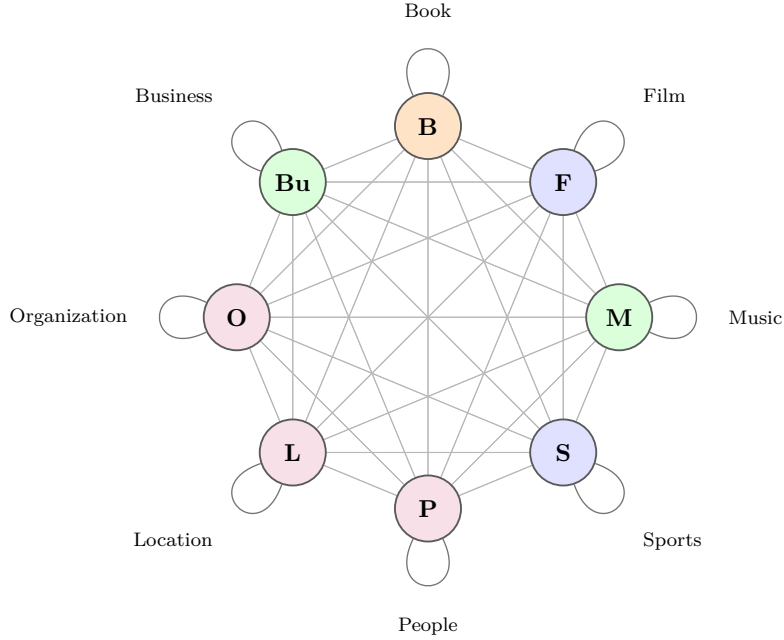


ogbn-mag / MAG240M



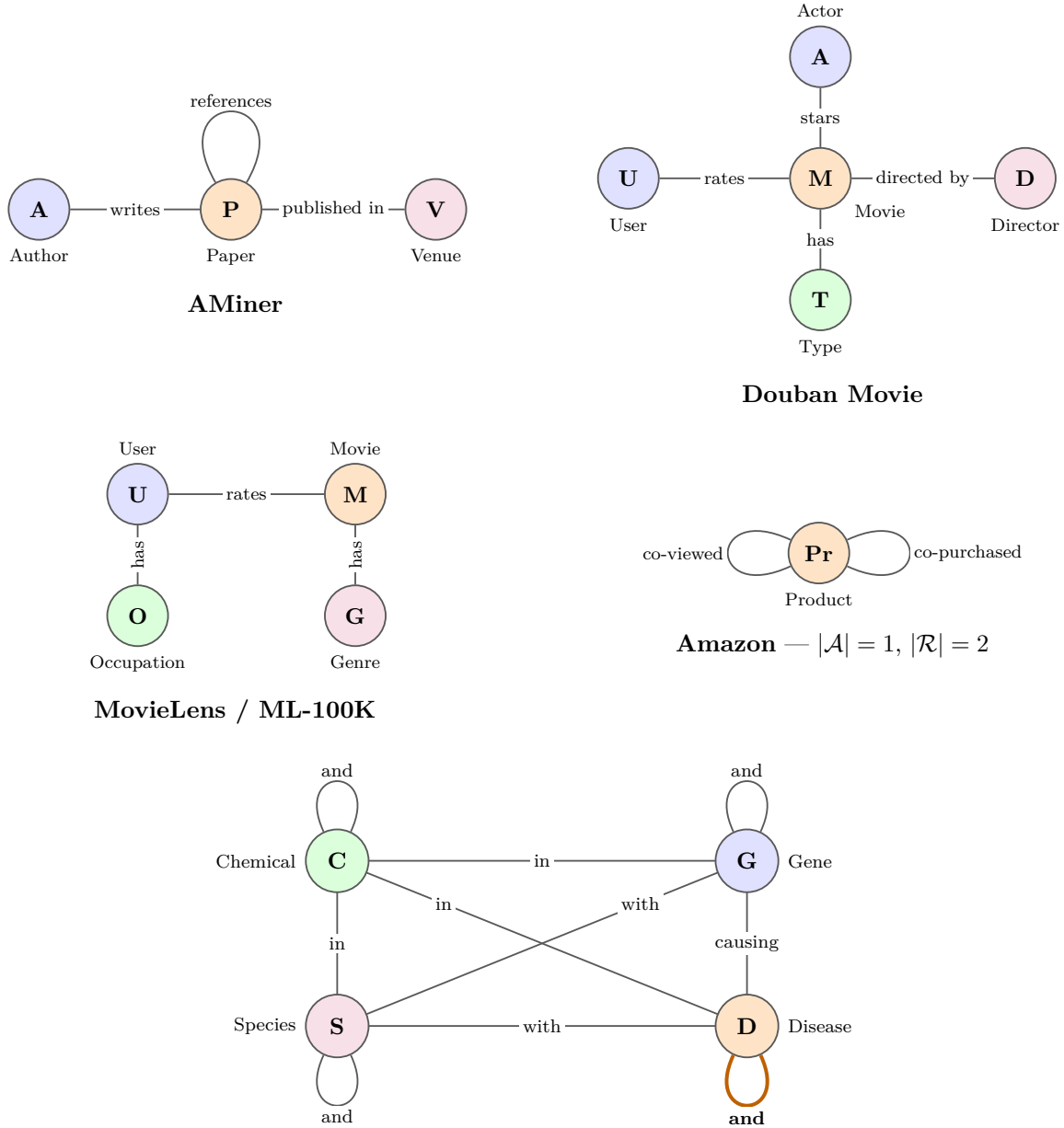
Yelp

Figure 1: Network schemas $T_G = (\mathcal{A}, \mathcal{R})$ of the bibliographic, media, large-scale and review benchmarks. Circles are node types (the letter is the initial used in meta-path notation); labelled edges are relation types. Inverse relations are omitted in this figure — but see Figure 2, where nothing is omitted.



Freebase (HGB) — the schema is *complete*: $\binom{8}{2} = 28$ pairwise relations + one self-relation per type = 36.

Figure 2: Network schema of Freebase (HGB), drawn *in full* — all 36 relation types are shown, so unlike Figure 1 nothing is omitted here. Freebase is the one benchmark whose schema is **complete**: every one of the $\binom{8}{2} = 28$ unordered type pairs carries exactly one relation, and every type carries one self-relation, giving $28 + 8 = 36$. Note the contrast with DBLP/ACM/IMDB: **none of these 36 are inverses** — all 36 are distinct forward relations, so this count is *not* inflated by the forward/inverse doubling of Section 1, whereas DBLP’s “6 edge types” is 3 relations plus their 3 inverses. BOOK (shaded) is the labelled target type; the largest single relation is MUSIC–MUSIC, 283,670 edges or 27% of the graph.



PubMed (HGB) — the schema is *complete*: $\binom{4}{2} = 6$ pairwise relations + one self-relation per type = 10. The bold loop is the link-prediction target, DISEASE–DISEASE.

Figure 3: Network schemas of the five benchmarks described in Section 2 but not drawn in Figures 1–2. Conventions as before; inverse relations are omitted. AMiner is drawn in the metapath2vec form (author/paper/venue), with the paper–paper reference relation that the HeCo subset also carries. Douban Movie and MovieLens/ML-100K are user–item HINs: the user–item rating edge is the prediction target, and the remaining types are side information attached to one side or the other. Amazon has a single node type, so its whole schema is one circle with two self-relations. PubMed is drawn *in full*: its four types are GENE, DISEASE, CHEMICAL, SPECIES and its ten relations are GENE-AND-GENE, GENE-CAUSING-DISEASE, DISEASE-AND-DISEASE, CHEMICAL-IN-GENE, CHEMICAL-IN-DISEASE, CHEMICAL-AND-CHEMICAL, CHEMICAL-IN-SPECIES, SPECIES-WITH-GENE, SPECIES-WITH-DISEASE and SPECIES-AND-SPECIES (edge labels abbreviate the verb); these are taken verbatim from the dataset’s own `info.dat`, and their counts sum to the published 244,986.

- **DeepWalk** — Perozzi, Al-Rfou & Skiena, KDD 2014. Truncated uniform random walks treated as sentences, fed to skip-gram with hierarchical softmax.
- **LINE** — Tang, Qu, Wang, Zhang, Yan & Mei, WWW 2015. Explicitly preserves first-order (observed edge) and second-order (shared-neighbourhood) proximity, optimised with negative sampling and edge-sampling for large graphs.
- **node2vec** — Grover & Leskovec, KDD 2016. DeepWalk with a biased second-order walk whose p, q parameters interpolate between BFS-like (structural equivalence) and DFS-like (homophily) exploration.
- **GCN** — Kipf & Welling, ICLR 2017. First-order spectral convolution: a layer is a symmetrically normalised neighbourhood average followed by a linear map and nonlinearity.
- **GAT** — Veličković, Cucurull, Casanova, Romero, Liò & Bengio, ICLR **2018**. Replaces the fixed normalisation with multi-head self-attention over neighbours. (Commonly miscited as 2017, the arXiv posting date.)
- **GraphSAGE** — Hamilton, Ying & Leskovec, NIPS 2017. Inductive framework that samples a fixed number of neighbours and aggregates with mean/LSTM/max-pool, enabling embeddings for unseen nodes.

4.2 Meta-path and random-walk HIN embedding

- **metapath2vec** — Dong, Chawla & Swami, KDD 2017. Random walks are *meta-path-guided*: at each step the walker must move to the node type prescribed by the meta-path scheme (e.g. APVPA), so the generated context is type-balanced instead of being dominated by the most frequent type. The walks feed a heterogeneous skip-gram. Note that the base model still normalises negative sampling over *all* nodes regardless of type; it is the **metapath2vec++** variant that normalises the softmax per node type.
- **HIN2Vec** — Fu, Lee & Lei, CIKM 2017. Rather than fixing meta-paths in advance, it learns node *and* relation (meta-path) vectors jointly by training a multi-task binary classifier that predicts, for a node pair, which relations hold between them.
- **ESim** — Shang, Qu, Liu, Kaplan, Han & Peng, arXiv:1610.09769, 2016 (*preprint only — never published in a peer-reviewed venue*). Embeds nodes by sampling positive and negative path instances of user-supplied weighted meta-paths and optimising a noise-contrastive objective for similarity search.
- **HERec** — Shi, Hu, Zhao & Yu, IEEE TKDE 31(2):357–370, **2019**. Performs meta-path-guided random walks, but first *filters* each walk to the target type, producing homogeneous sequences per meta-path; the resulting embeddings are fused by a nonlinear function and injected into an extended matrix-factorisation recommender.
- **PTE** — Tang, Qu & Mei, KDD 2015. Decomposes a heterogeneous text network (word–word, word–document, word–label) into bipartite graphs and jointly embeds them with the LINE objective, giving semi-supervised, task-tuned text embeddings.

4.3 Heterogeneous graph neural networks

- **R-GCN** — Schlichtkrull, Kipf, Bloem, van den Berg, Titov & Welling, ESWC **2018**. Relation-specific message passing: each layer aggregates neighbours separately per relation r with its

own weight matrix W_r , normalised by $c_{i,r} = |\mathcal{N}_i^r|$, plus a self-loop term. Because $|\mathcal{R}|$ weight matrices overfit, it offers two *alternative* regularisers — *basis decomposition* (W_r a linear combination of B shared bases) and *block-diagonal decomposition* (W_r a direct sum of small dense blocks). For link prediction it is used as an encoder with a **DistMult** decoder.

- **HAN** — Wang, Ji, Shi, Wang, Ye, Cui & Yu, WWW 2019. Two-level attention: *node-level* GAT-style attention within each meta-path-induced neighbour set, then *semantic-level* attention learning a scalar importance per meta-path to fuse them. Its limitation — the one MAGNN attacks — is that it operates on meta-path-*induced* graphs, so intermediate nodes along the meta-path are discarded.
- **MAGNN** — Fu, Zhang, Meng & King, WWW 2020. Three stages: node-content transformation projecting differently-dimensioned type attributes into a shared space; *intra*-meta-path aggregation that encodes the **entire path instance including intermediate nodes** (mean, linear, or a relational-rotation encoder borrowed from RotatE) with attention over instances; and *inter*-meta-path semantic attention.
- **GTN** — Yun, Jeong, Kim, Kang & Kim, NeurIPS 2019. *Learns* meta-paths instead of accepting them. A 1×1 convolution with softmax weights forms a soft convex combination Q_i of the per-edge-type adjacency matrices; multiplying $A = Q_1 Q_2 \cdots Q_l$ yields new meta-path adjacencies differentially. Including the identity among the candidates lets it learn paths *shorter* than l . A GCN then runs on the learned graph.
- **HetGNN** — Zhang, Song, Huang, Swami & Chawla, KDD 2019. Restart-based random walks sample a fixed-size, type-balanced neighbour set; content of each node is encoded with a Bi-LSTM over its heterogeneous attributes, neighbours are aggregated per type, and types are combined by attention.
- **HGT** — Hu, Dong, Wang & Sun, WWW 2020. Node-type-specific projections produce Key/Query/Value, and attention and message matrices are indexed by the *meta-relation* $\langle \tau(s), \phi(e), \tau(t) \rangle$, with a learnable scalar prior μ per meta-relation. It needs no hand-designed meta-paths. Adds *relative temporal encoding* for dynamic graphs and *HGSampling* for web-scale mini-batch training.
- **simple-HGN / HGB** — Lv, Ding, Liu et al., KDD 2021. A deliberately minimal baseline: GAT plus (i) a learnable **edge-type embedding** concatenated into the attention logit, (ii) residual connections on *both* node representations and attention scores, and (iii) L_2 normalisation of the output embeddings. The paper’s larger contribution is the HGB benchmark itself and the finding that this simple model matches or beats most specialised HGNNs.
- **ie-HGCN** — Yang, Guan, Li, Zhao, Cui & Wang, IEEE TKDE 35(2):1637–1650, 2023. Hierarchical object-level then type-level aggregation which, as a by-product, *ranks* the useful meta-paths, making the model interpretable while avoiding an exhaustive meta-path search.
- **RSHN** — Zhu, Zhou, Pan, Zhu & Wang, IEEE ICDM 2019. Builds a coarsened *edge-centric* “relation structure graph” to embed relation types themselves, then propagates with those relation embeddings rather than treating relations as bare labels.
- **HPN** — Ji, Wang, Shi, Wang & Yu, IEEE TKDE 35(1):521–532, 2023. Analyses semantic confusion (the HIN analogue of over-smoothing) and replaces the propagation step with a personalised-PageRank-like update that retains a fraction of the node’s own representation, enabling deeper meta-path propagation.

4.4 Self-supervised / contrastive heterogeneous methods

- **DMGI** — Park, Kim, Han & Yu, AAAI 2020. Extends Deep Graph Infomax to *attributed multiplex* networks: one DGI mutual-information objective per relation type, with a consensus regularisation that pulls the per-relation embeddings towards a shared representation.
- **HDGI** — Ren, Liu, Huang, Dai, Bo & Zhang, arXiv:1911.08538, 2019 (AAAI-20 DLGMA workshop, under a different title; *no archival venue*). Applies mutual-information maximisation between local node embeddings and a global summary, using meta-path-based convolution plus semantic attention as the encoder.
- **HeCo** — Wang, Liu, Han & Shi, KDD 2021. Contrasts two *views* of the same node: a *network-schema* view (attention over directly-connected neighbours of each other type) and a *meta-path* view (meta-path-specific GCNs with semantic attention). Positives are not merely the node’s counterpart view — HeCo selects multiple positives by counting how many meta-path instances connect two nodes and taking the top ones.

4.5 Knowledge-graph embeddings as relation-aware LP baselines

A HIN with many relation types is close to a knowledge graph, so heterogeneous link-prediction papers (and HGB’s LP track) routinely report KG embedding models, which score a typed triple (h, r, t) directly.

- **TransE** — Bordes, Usunier, Garcia-Duran, Weston & Yakhnenko, NIPS 2013. Models a relation as a translation, $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$, scoring $-\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|$. Cannot represent 1-to-N or symmetric relations.
- **DistMult** — Yang, Yih, He, Gao & Deng, ICLR 2015. Bilinear score $\langle \mathbf{h}, \text{diag}(\mathbf{r}), \mathbf{t} \rangle$; efficient but necessarily *symmetric*.
- **Complex** — Trouillon, Welbl, Riedel, Gaussier & Bouchard, ICML 2016. Moves DistMult into $\mathbb{C}$ and takes the real part of a Hermitian product, so asymmetric relations become representable at the same cost.
- **ConvE** — Dettmers, Minervini, Stenetorp & Riedel, AAAI 2018. Reshapes h and r into 2-D and applies a convolution, giving a parameter-efficient, highly expressive scorer.
- **RotatE** — Sun, Deng, Nie & Tang, ICLR 2019. Treats a relation as an element-wise *rotation* in complex space, $\mathbf{t} = \mathbf{h} \circ \mathbf{r}$ with $|r_i| = 1$, capturing symmetry, antisymmetry, inversion and composition simultaneously.

5. Evaluation metrics

Two conventions govern the whole literature. First, a HIN benchmark normally labels **exactly one node type** — the author in DBLP, the paper in ACM, the movie in IMDB — so “accuracy on DBLP” always means accuracy on that one target type, the other types being context. Second, for *unsupervised* methods the protocol is **embed** $\rightarrow$ **freeze** $\rightarrow$ **fit a linear classifier**: the embedding is learned without labels, then held fixed while a logistic regression is trained on a fraction of target nodes, the fraction varied over 20/40/60/80% and mean $\pm$ std reported over repeated splits. HAN, MAGNN and HeCo all report in this form.

5.1 Node classification

With class set $\mathcal{C}$ and per-class counts $\text{TP}_c, \text{FP}_c, \text{FN}_c$, **Micro-F1** pools the confusion matrix *before* scoring,

$$\begin{aligned}\text{Micro-P} &= \frac{\sum_c \text{TP}_c}{\sum_c (\text{TP}_c + \text{FP}_c)}, & \text{Micro-R} &= \frac{\sum_c \text{TP}_c}{\sum_c (\text{TP}_c + \text{FN}_c)}, \\ \text{Micro-F1} &= \frac{2 \text{Micro-P} \cdot \text{Micro-R}}{\text{Micro-P} + \text{Micro-R}},\end{aligned}$$

whereas **Macro-F1** scores each class and averages *unweighted*:

$$\text{F1}_c = \frac{2 \text{TP}_c}{2 \text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{Macro-F1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \text{F1}_c.$$

In the *single-label* multi-class setting every misclassification creates one false positive and one false negative, so $\text{Micro-P} = \text{Micro-R}$ and Micro-F1 equals plain **accuracy**. The pair is always reported together because they answer different questions: Micro-F1 is dominated by frequent classes, Macro-F1 weights a rare class as heavily as a common one, and a large gap between them signals a model that has learned the class prior rather than the graph.

The multi-label caveat. This equivalence **fails** for HGB’s IMDB and Freebase tracks, where a node may carry several labels at once. Prediction is per-label thresholding (sigmoid + binary cross-entropy) rather than an arg max, and the pooled confusion matrix is taken over the flattened node $\times$ label indicator matrix; there $\text{Micro-P} \neq \text{Micro-R}$ in general and Micro-F1 is *not* accuracy. In HGB’s IMDB, 2,684 of the 4,573 labelled movies carry more than one genre, so this is not a corner case — and it is why a Micro-F1 on HGB’s IMDB cannot be compared with one on MAGNN’s single-label IMDB.

Other classification metrics. **Accuracy** is the official metric on OGB (ogbn-mag, MAG240M). **ROC-AUC** is used for binary or strongly imbalanced targets (and by HeCo alongside F1). For the *clustering* probe that HAN, MAGNN, DMGI and HeCo all run — k -means on the frozen embeddings with $k = |\mathcal{C}|$ — the scores are

$$\text{NMI}(Y, \hat{C}) = \frac{2 I(Y; \hat{C})}{H(Y) + H(\hat{C})} \in [0, 1], \quad \text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]},$$

with I mutual information, H entropy and RI the Rand index. Both are 0 in expectation for a chance partition and 1 for a perfect one (ARI may go negative). These test whether the embedding geometry is class-structured with no classifier fitted at all.

5.2 Link prediction

The essential difference from the homogeneous case is that in a HIN link prediction is stated **per relation type**: not “is there an edge” but “is there an edge *of type R*” — predict Author–Paper, or Paper–Venue, or User–Artist. Two consequences follow. (i) *Negatives must be sampled type-consistently*: a negative for an Author–Paper edge must join an author to a paper, never an author to a venue, since a type-violating negative is trivially rejected by any type-aware model and inflates its apparent margin over type-agnostic baselines. HGB samples negatives from 2-hop neighbours at a 1:1 ratio for exactly this reason. (ii) Scores are reported *per relation* and then averaged, so

a model can be strong on a dense relation and useless on a sparse one without that showing in a single number.

Let rank_i be the position of the i -th true edge among its candidates, over query set Q :

$$\text{MRR} = \frac{1}{|Q|} \sum_{i \in Q} \frac{1}{\text{rank}_i}, \quad \text{Hits@K} = \frac{1}{|Q|} \sum_{i \in Q} \mathbf{1}[\text{rank}_i \leq K],$$

i.e. MRR is the mean reciprocal rank of the first true item and Hits@ K the fraction of positive edges ranked inside the top K against their sampled negatives ($K \in \{1, 3, 10\}$ conventionally, inherited from the KG literature). **ROC-AUC** equals the probability that a random positive outranks a random negative. **Average precision** (equivalently AUPRC) is

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n,$$

with P_n, R_n precision and recall at the n -th threshold. AP is preferred to ROC-AUC when positives are rare — always true for link prediction on a sparse graph — because ROC-AUC is optimistic there, the enormous true-negative count sitting in its denominator. **Precision@ K / Recall@ K** , and MAE/RMSE for rating prediction, appear in the recommendation-flavoured HIN papers (HERec on Yelp/Douban). HGB’s LP track standardises on **ROC-AUC and MRR**.

5.3 Summary: dataset $\rightarrow$ target $\rightarrow$ task $\rightarrow$ metric

Which metric is “correct” is not a property of the task in the abstract but of the dataset–task pair, because it is the dataset that fixes the target node type, whether the labels are single- or multi-label, and how many classes there are. Table 5 therefore closes the section by mapping each benchmark surveyed in Section 2 to its target type, its task, and the metrics that paper after paper actually reports on it. Two entries in it are worth singling out: the OGB datasets break the Macro-/Micro-F1 habit of the HIN literature and are scored by the leaderboard on plain **accuracy**, and the recommendation-flavoured datasets (Yelp, Douban) are scored on **MAE/RMSE** because their task is rating prediction rather than classification.

Table 5: Dataset $\rightarrow$ target node type $\rightarrow$ task $\rightarrow$ metric normally reported.

Dataset	Domain	Target	Task	Labels	Metrics normally reported
DBLP	bibliographic	author	NC	4, single	Macro-F1, Micro-F1 (+NMI/ARI)
ACM	bibliographic	paper	NC	3, single	Macro-F1, Micro-F1 (+NMI/ARI)
IMDB (HGB)	movie	movie	NC	5, multi	Macro-F1, Micro-F1
IMDB (MAGNN)	movie	movie	NC	3, single	Macro-F1, Micro-F1
Freebase (HGB)	KG	book	NC	7, single	Macro-F1, Micro-F1
AMiner	bibliographic	author/paper	NC	8 / 4	Macro-F1, Micro-F1, NMI
ogbn-mag	bibliographic	paper	NC	349, single	Accuracy (official)
MAG240M	bibliographic	paper	NC	153, single	Accuracy (official)
LastFM	music	user–artist	LP	—	ROC-AUC, MRR
Amazon	e-commerce	product–product	LP	—	ROC-AUC, MRR
PubMed (HGB)	biomedical	disease–disease	LP	—	ROC-AUC, MRR
Yelp / Douban	review	user–item	LP / rating	—	Prec@K, Rec@K; MAE, RMSE
OAG (HGT)	bibliographic	paper–field/venue	LP	—	NDCG, MRR

Closing remark. The single most important methodological point is the one HGB was created to make: because DBLP, ACM and IMDB circulate in mutually incompatible preprocessed versions (Table 3), and because IMDB even changes from single-label to multi-label between versions,

numbers copied across papers are frequently not comparable. Any credible evaluation must name the dataset *version*, the split, and whether the protocol is end-to-end supervised or embed-freeze-classify.